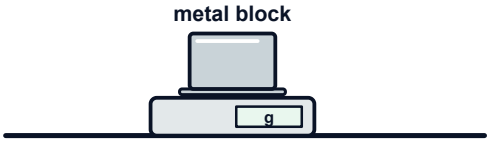
DO NOW · REQUIRED PRACTICAL

# Specific heat capacity

**Energy**
Paper 1 · Combined and Separate
6 steps

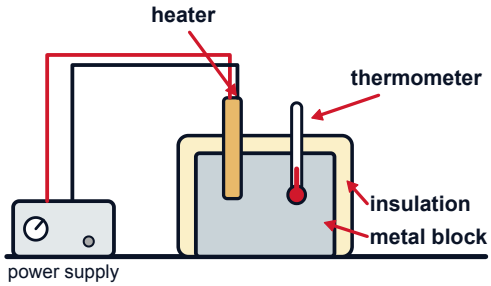
A student measures the specific heat capacity of a metal block.

## 1 Measure the mass of the block with a balance.

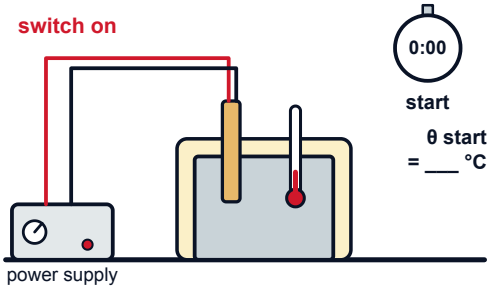



measure its mass m

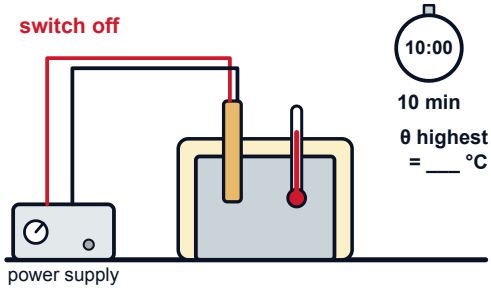
## 2 Put a heater and a thermometer into the block and wrap it in insulation.




## 3 Record the starting temperature, then switch on the heater and start a timer.




## 4 After 10 minutes, switch off and record the highest temperature reached.




## 5 Calculate the energy supplied from the power of the heater and the time.

$$E = P \times t$$

energy (J) = power (W) × time (s)

time in seconds:
**10 minutes = 600 s**

## 6 Calculate the specific heat capacity: energy ÷ (mass × temperature rise).

$$c = E \div (m \times \Delta\theta)$$

J/kg°C = J ÷ (kg × °C)

**Δθ = θ highest − θ start**
m in kg (not g)

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